Ibrahim Ali

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Education

- **Vellore Institute of Technology** — Chennai, India
  B.Tech in Computer Science and Engineering; CGPA: 8.7 — *July 2020 – May 2024*

Experience

- **Air India Limited** — Kochi, India
  Software Engineer — *August 2024 – Present*
  - Backend engineer building large-scale event-driven passenger communication platforms supporting airline operations.
  - Owned and scaled a real-time event-driven delivery platform processing ∼**3,500+ flight events/day** and delivering ∼**500,000 passenger notifications/day**, replacing a polling-based scheduler that generated ∼**30,000 daily API calls to the Amadeus reservation system** with an event-driven model requiring only one call per event, significantly reducing external system load.
  - Optimized the notification processing architecture, reducing operational costs from ***Rs.*40L/month to under *Rs.*10L/month** through improved event processing efficiency, channel optimization, and system consolidation.
  - Migrated Air India's notification delivery platform to an **event-driven architecture** using **Azure Event Hub and Service Bus**, scaling the system to handle ∼**1M+ notifications/day** with peak loads exceeding **50,000 requests/hour**.
  - Architected a distributed scheduling engine orchestrating time-based notification workflows for **100,000+ booking events/day** and **400,000+ scheduled notifications**, combining event-driven booking ingestion and periodic flight schedule evaluation to trigger configurable notification pipelines.
  - Implemented a configuration-driven rule engine supporting dynamic trigger offsets, flight/route targeting, passenger segmentation, and quiet-hour aware scheduling, enabling reliable notification delivery through asynchronous pipelines with built-in retry and recovery mechanisms.
  - Built an **event-driven flight schedule change processing system** that ingests reservation updates, evaluates delay and prepone scenarios through rule-based validation, and automatically triggers passenger notifications, eliminating a multi-team manual workflow and reducing operational costs by ∼**40%**.
  - Engineered a scalable survey distribution pipeline processing **150,000+ burst records** through **Azure Event Hub**, introducing **Azure Data Factory**-based orchestration to regulate third-party API throughput and **Service Bus**-driven asynchronous callbacks to reliably process generated survey URLs and dispatch customer notifications with retryable state tracking.
  - Introduced a **WhatsApp & Email-first notification strategy**, reducing SMS and robocall costs by ∼**60% per month** while improving delivery reliability and customer engagement.
  - Designed and developed an internal **MCP client & server** with tool-based access to backend **Spring Boot** services, enabling secure on-demand database queries and API calls to retrieve live production system state and reducing investigation time from hours to minutes.
  - **Multiple Excellence Awards (2024–2026)** — Received **Dream Team Award (1x)**, **High Flyer Award (2x)**, and **AI IMPACCT Champion Award (2x)** for driving system reliability, production stability, and operational optimization across critical passenger communication platforms.
  - Maintained production-grade reliability through structured **PR reviews**, **SonarQube analysis**, automated unit testing, and **GitHub security scanning**, while implementing **Grafana dashboards** for real-time system monitoring and streaming finalized notification records to **Databricks** for analytics and operational insights.

Projects

- **Flight Disruption Management System (Prototype)**: — Python, React.js, Claude Code, LangChain, PSQL
  - Designed an initial **4-agent architecture** (analysis, severity scoring, option generation, decisioning) for disruption impact evaluation and resolution generation.
  - Independently enhanced and production-hardened the system by introducing **deterministic state transitions** across **5 conversation stages**, improving reliability and traceability.
  - Expanded agent coordination logic and memory handling, reducing manual intervention during disruption spikes and improving response consistency.

Skills

- Languages: Java, C/C++, Python, SQL, JavaScript
- Backend: Spring Boot, REST APIs, SOAP APIs, Microservices, Event-Driven Systems
- Distributed Systems: Messaging, Fault Tolerance, Workflow Orchestration
- Cloud Platforms: Azure, AWS (S3, EC2, Lambda, CloudWatch)
- Containerization & CI/CD: Docker, Kubernetes, Azure Pipelines
- Databases: PostgreSQL, MySQL, CosmosDB, Redis
- Foundations: Object-Oriented Design, Design Patterns, Data Structures